भारत सरकार
Govt. of India
श्रम एवं रोजगार मंत्रालय
Ministry of Labour & Employment
खान सुरक्षा महानिदेशालय
Directorate-General of Mines Safety

NO: 150153|SEZ|Raigarh Region|Perm|2022|238365 Date: 30/05/2022

To

The Agent,

Samri Bauxite Mine

M/s Hindalco Industries Ltd.

Sub: Permission under Reg.106(2)(b) of the Metalliferous Mines Regulations, 1961 for deployment of Heavy Earth Moving Machinery for digging, excavation and removal of overburden and bauxite in conjunction with deep hole blasting at Samri Bauxite Mine of M/s Hindalco Industries Limited.

Sir,

Please refer to your online application ID No.238365 dated 19.04.2022 and other offline/physical documents, Plans & sections submitted on the above subject.

The matter has since been examined on the basis of the information furnished and shown on the accompanied plans submitted by you. In exercise of the powers conferred on the Chief Inspector of Mines (also designated as Director-General of Mines Safety) under the provisions of Regulations 106(2)(b) of the Metalliferous Mines Regulations, 1961 and by virtue of authorization granted to me by the Chief Inspector of Mines (also designated as Director General of Mines Safety) under Section 6(1) of the Mines Act, 1952, I hereby permit you to deploy Heavy Earth Moving Machinery for digging, excavation & removal of overburden and bauxite in conjunction with deep hole blasting at Kudag Bauxite Mine of M/s Hindalco Industries Limited, in the areas bounded by **(1)** A1-A2…..A7-A8 & A1; **(2)** B1-B2…..B51-B52 & B1; **(3)** C1-C2…..C9-C10 & C1; **(4)** D1-D2….D8-D9 & D1; **(5)** E1-E2….E35-E36 & E1; **(6)** F1-F2….F43-F44 & F1; **(7)** G1-G2….G19-G20 & G1; **(8)** H1-H2….H27-H28 & H1; **(9)** I1-I2….I118-I119 & I1; **(10)** J1-J2….J16-J17 & J1; **(11)** K1-K2…K15-K16 & K1; **(12)** L1-L2….L17-

L18 & L1; **(13)** M1-M2……M15-M16 & M1; **(14)** N1-N2…….N7-N8 & N1; **(15)** O1-O2……O16-O17 & O1; **(16)** P1-P2……O48-O49 & O1; **(17)** P1-P2…..P48-P49 & P1; **(18)** Q1-Q2……..Q12-Q13 & Q1; **(19)** R1-R2…….R9-R10 & R1; **(20)** S1-S2…….S22-S23 & S1; **(21)** T1-T2…….T15-T16; **(15)** U1-U2…..U16-U17 & U1; **(16)** V1-V2……V29-V30 & V1; **(17)** W1-W2……W36-W-37 & W1; **(18)** X1-X2……X48-X49 & X1 as shown on plan No.HIL/SBM/SP/2022/01 dated 31.03.2022 (Part-1, 2, 3, 4, 5, 6, 7 and Part-8), subject to the following conditions being strictly complied with:

(1) A person holding a First Class Managers' Certificate of Competency under the Metalliferous Mines Regulations, 1961, shall remain as Manager of the mine. This permission shall stand revoked as soon as the qualified Manager ceases to work at the mine.

(2) Adequate numbers of supervisors including qualified Foremen & Mates shall be appointed to assist the manager. Mine Foreman's district shall not contain more than two excavators with ancillary equipment viz. dumpers, loaders and drills etc and one Foreman shall not supervise work of more than two Mining Mates.

(3) No ore dressing/ handling/ processing plant shall be attached with the mine.

(4) No person shall be engaged in ore/ overburden benches within 60m of any Heavy Earth Moving Machinery deployed therein.

(5) Adequate general lighting as required under notification No. GSR-618(E), dated 28th April, 2017, published in the Gazette of India dated 21st June, 2017, Part II Section 3(i) (DGMS Legis. Circular No.03 of 2017) shall be kept provided and maintained during working hours at different places or areas where natural light becomes insufficient.

(6) Drilling, Charging and Blasting operation shall be done under the personnel supervision of the Mine Manager or a separate Officer holding at least a Second Class Managers' Certificate of Competency under the Metalliferous Mines Regulations, 1961.

(7) Blasting parameters with a rough sketch showing the drilling pattern and the holes charged shall be maintained in a register kept for the purpose for each blast.

(8) No work of removal of overburden and extraction of ore by use of explosives shall be done within 300m of surface structure/ buildings unless separate permission under Regulation 164(1-B) of the Metalliferous Mines Regulations, 1961 is obtained from this Directorate.

(9) No working with Heavy Earth Moving Machinery shall be extended within 45.0m of any surface feature or public road not belonging to the owner unless the same is acquired or consent to operate is obtained from the owner concerned.

(10) A suitable 'Code of Traffic Rules' for regulating the movement of Heavy Earth Moving Machinery (commensurate with the capacity/ size type of machines used in the mechanised opencast mine) shall be framed and enforced immediately. Such Code of Traffic Rules should be approved by Tri-partite Committee.

(11) The operator's cabins of Heavy Earth Moving Machinery shall be well designed and substantially built so as to ensure adequate protection to the operator against heat, dust, noise etc. and at the same time provide adequate safety to the operator in the event of overturning of Heavy Earth Moving Machinery. A seat belt for the safety of the operator shall also be provided.

(12) A code of practice shall be drawn up for dealing with fires at different locations in the opencast mine, including HEMM. Arrangements for fighting fire should be provided on all Heavy Earth Moving Machineries. Such arrangements should, if possible, operate automatically on appearance of fire.

(13) Separate ore stockyards shall be maintained for dumping and dispatch of ore and it shall be ensured that dispatch of mineral is not carried out from the dump yard where dumping is under progress. Further, buyers'

trucks shall be parked away from the mine premises and only such trucks shall be allowed to enter the mine, which are due for loading during the shift. Also, it shall be ensured that only permitted quantity of mineral is loaded in the trucks so as to obviate the risk of accidents due to overloaded trucks in the mine premises.

(14) All persons to be employed to drive/ operate HEMM should be trained and their competency should be evaluated by a Board constituted by the management. The members of such board shall be persons who are not connected with imparting of training. However, the training officers may be co-opted in the Board as observer.

(15) Only such fitters/ mechanics who possess driver's/ operator's license should be allowed to carry out test-run of HEMM.

(16) The portion of surface haul road in mine premises where there is heavy traffic of men and machines should be provided with a separate lane properly fenced off from the haul road for pedestrians and two wheelers.

(17) Trucks and other heavy vehicles, not belonging to management should not be allowed in the mine premises without a valid pass issued by the manager or other competent person. Before the pass is issued the engineer shall check the road-worthiness of such vehicles.

(18) In order to check entry of unauthorized vehicles into the mine premises, a properly manned check gate(s) at the entrance(s) shall be set up where record of entry and exit of each such vehicle shall be maintained.

(19) At the check gate the license of the drivers shall also be checked for eliminating the possibility of unlicensed persons driving the vehicles.

(20) Persons engaged in surface operations and, in particular, the contractor's workers, who incidentally are often inexperienced and least informed about job-safety matters, need closer and more competent supervision. To minimize accidents due to surface operations it shall be ensured that all persons engaged at any work within the mine premises through the contractors have received relevant training and other job-related briefings and that the drivers of vehicles belonging to contractors entering the mine premises have additionally been explained the salient provisions of "traffic rules".

(21) Each and every operation, including the operation carried out through contractor's workers or by outside agency, shall be placed under the charge of a competent supervisor, duly appointed and authorized by the manager, his jurisdiction being clearly demarcated.

(22) (a) A code specifying the duties and responsibilities of all officers of mining cadre, Mine Engineer, Supervisors, Technicians, Mechanics, Fitters, Machine Operators, helpers, loading supervisors etc. shall be drawn up and distributed to all.

(b) Safety management plan shall be developed by mine management under guidelines issued vide Circular(tech) 03 of 2019. Hazard identification and risk management shall be carried out with participation of departmental and contractual work persons. The mine Manager shall submit a copy of safety management plan to RIM.

(c) It shall be the responsibility of the Manager, Engineer and other supervisors to see that all persons working in the mine in all machines/equipments etc. work as per the code and all machines and equipments etc. are installed, operated and maintained in safe working conditions.

(23) In the event of any change in the circumstances connected with this permission which is likely to endanger the life of persons employed in the mine or the mining operations for which this permission has been granted shall be stopped forthwith and intimation sent to the Director of Mines Safety thereof. The said mining operation shall not be resumed without express and fresh permission in writing from this Directorate.

(24) If at any time any one of the conditions subject to which this permission has been granted is violated or not complied with, this permission shall be deemed to have been revoked with immediate effect.

(25) This permission may be amended or withdrawn at any time if considered necessary in the interest of safety and is being issued under Regulation 106(2)(b) of the Metalliferous Mines Regulations, 1961 only without prejudice to any other provisions of law which may be or may become applicable at any time.

(26) This Directorate shall be informed as soon as the mining operations are commenced in accordance with the above permission. Intimation about completion of the mining operations should also be sent promptly and in any case not later than one month thereof.

(27) This permission shall be valid for a period of five (5) years from the date of issue of this letter.

Annexure 'A'

1.0. GENERAL:

I. Except where otherwise provided for in this conditional permission all provisions of the Metalliferous Mines Regulations, 1961 relating to opencast workings, use of explosives and machinery shall be strictly complied with.

II. (a) All damages to property not belonging to the owner of the mine and injuries to persons as a result of mining operations shall be adequately indemnified. Should there be any controversy in this respect; the decision of the DGMS shall be final and binding on the management.

(b) No workings shall be extended to any point within 45m of village, dwellings, Roadway, Roadway, Railway acquired land and Public structure, H.T. Power line etc.

III. No work whatsoever shall be done where the provisions of Regulation 126 of the Metalliferous Mines Regulations, 1961 are attracted due to the presence of River, jore and nallah in the vicinity. The entire ground lying within 15m of all the River, nallahs and jores shall be filled up and raised and consolidated to a R.L. of at least 3m above the highest flood level.

IV. Safety management plans shall be prepared after proper Hazard identification as per guidlines mentioned in DGMS(Tech) Circular No. 03 of 2019.

2.0. OPENCAST WORKINGS:

(a) The height of the benches in alluvium and soft rock formation shall not be more than 3m and in overburden, ore or other hard rock formation not be more than the digging height of the machine used for digging, excavation or removal.

(b) The width of any bench shall not be less than: (i) the width of the widest machine plying on the bench plus two meters, or (ii) if dumpers ply on the bench three times the width of the dumper, or (iii) the height of the bench, whichever is more.

(c) If tippers or similar transport vehicles are deployed for transportation of ore, separate roads for such tipping trucks or similar vehicles shall be provided and safe loading practices shall be adopted avoiding plying of trucks on the same bench where men are to work, travel or rest.

(d) No manual workers shall be employed where HEMM is deployed in the mine.

3.0. FENCING AROUND OPENCAST WORKINGS:

(a) The periphery around the limits of opencast workings shall be kept fenced with a masonry wall not less than 0.40m thick and not less than 1.2m high, with parapet top.

(b) The edges of benches of the opencast workings shall be kept fenced with wire ropes strand or barbed wire supported by movable posts (wooden, iron or concrete). The gap between the adjacent rope strands or wires shall not be more than 0.30m and the bottom most rope strand wire shall not be more than 0.25m and the top most rope strand or wire not less than 1.0m from ground level.

(c) Before abandoning the mine, the pits/ quarry shall be filled upto adjacent ground level by earth, where ever applicable.

4.0. SPOIL OVERBURDEN OR DEBRIS BANKS:

I (a) Spoils, overburden or debris shall be deposited at places belonging to the mine and duly approved by the manager in writing.

b) Spoils, overburden or debris shall not be deposited, beneath transmission, telephone or power lines or near any public structure.

(c) The slope of a spoil bank face shall be determined by natural angle of repose of the material being deposited, but shall in no case exceed 37.5 degree from the horizontal. The spoil bank face shall not be retained by artificial means at an angle in excess of its natural angle of repose.

(d) Any spoil bank exceeding 30m in height shall be kept benched so that no such bench exceeds 30m in height and the general slope does not exceed 1 vertical to 1.5 horizontal. In no case, the height of any spoil bank shall be allowed to exceed 60m unless a scientific study on its stability has been done and a separate permission is obtained from this Directorate.

(e) The spoil bank face shall not be retained by and artificial means at an angle in excess of its natural angle of slope.

(f) DGMS(Tech) circular No. 02 of 2020 dated 09.01.2020 shall be complied for systematic slope monitoring of benches in opencast workings or overburden dumps.

II. (a) The spoils overburden or debris shall not be deposited within 45m of a railway or other structures and it shall not be permitted to approach any of the above said structures closer than a distance equal to 1.5 times the vertical height of its face.

(b) The toe of a spoil-bank shall not be permitted to approach a railway or other public works, public road or building or other permanent structure not belonging to the owner of the mine closer than 100m.

(c) A suitable fence shall be erected between any railways of public works or road or building or structures and the toe of an active spoil bank so as to prevent un-authorised persons from approaching the spoil bank.

(d) The backfilled area shall be kept benched and the distance of any active mine workings (faces) from the toe of bottom-most backfilled face (bench) shall not be less than 100m.

III. No person shall, or shall be permitted to approach the toe of an active spoil bank where he may be endangered from material rolling down the face. Suitable signs for warning the persons at conspicuous places shall also be displayed.

IV. Dozers shall be provided with the following safety features in addition:

1) Roll over protection

2) Turbo charge guard.

3) Fire resistant hydraulic hoses and wiring near hot zone

4) Seat belt.

5.0. ROADS FOR TIPPING TRUCKS AND DUMPERS ETC:

I. All haul roads for dumpers or other mobile machinery shall be constructed to suit its load capacity and maintained in good condition. No vehicles other than heavy transport vehicles shall use haul roads except between designated points and when permitted in writing by the manager.

II. Where practicable, all roads leading from the opencast workings to surface shall be arranged to provide one-way traffic and no road shall be of a width less than three times plus 5m the width of the largest vehicle plying on that road. Where it is not practicable, definite turnouts, crossing points and waiting points shall be designated for use of vehicle.

III. All corners and bends shall be made in such a way that the operators and drivers of vehicle have clear view of distance of not less than 3 times the breaking distance of largest HEMM working at 40 Km/hour.

IV. Where it is not possible to ensure a visibility for a distance as mention in (III) there shall be provided with two roads of width not less than 2 times plus 3m of largest vehicle plying on the road with a strong road divider at centre with adequate lighting and reflector along the divider.

V. Where any road existing above level of surrounding area, it shall be provided with strong parapet wall or embankment of following dimensions:

(a) Width at top not less than 1m.

(b) Width at bottom not less than 2.5m.

(c) The height not less than the diameter of any type of largest vehicle plying on road. It may be noted that just dumping of mud or OB shall not be treated as strong parapet wall.

VI. No road shall have gradient more than 1 in 16. Ramps with 1 in 10 gradients should not be more than 10m at one stretch.

VII. If large capacity dumpers and smaller capacity tipping trucks or similar transport vehicle are deployed in the mine, separate roads for tipping trucks or similar transport vehicles shall be provided.

VIII. Plying of dumpers or tipping trucks on the same bench where men are to work, travel or rest shall be avoided.

IX. Warning notices and road signs shall be posted along the haul roads at appropriate places like crossings, curves etc for guidance of drivers of dumpers/ trucks/ tippers.

X. At every curve, parapet wall or vertical posts with Zebra lines shall be provided to help the drivers to keep the trucks/ tippers/ dumpers on the track.

6.0. ADDITIONAL DUTIES OF ASSISTANT MANAGER INCHARGE OF OPENCAST WORKING:

I. During every shift, the opencast workings shall be placed under the charge of a qualified experienced Assistant Manager to have overall control and guidance of all operations.

II. During his shift the Assistant Manager shall:

a) Make frequent inspections for evidence of any unsafe conditions including evidence of slide of material from high wall and shall guide so that operations are conducted in a safe and efficient manner.

b) Not allow any person to work under overhanging edges or where there is evidence of slides, until such danger has been removed.

c) Ensure that every person engaged in dressing operations on high walls is provided with and uses safety belt of a type approved by this Directorate.

d) Ensure that all loose material is removed from high walls before drillers are engaged there; and

(e) Ensure that parapet walls along truck roads are properly maintained.

7.0 ADDITIONAL DUTIES OF ENGINEERS PLACED IN CHARGE OF MACHINES AND EQUIPMENTS IN OPENCAST WORKINGS:

The machines and equipments at work shall be placed under the charge of qualified and experienced engineer to effect inspection, examination, safe operations and maintenance of the machines, equipments and accessories. The engineer/engineers shall;

a) inspect, examine machines, equipments and accessories and satisfy himself that they are in sound and safe working order;

b) not allow any machine, equipment to be used, if it is found defective;

c) ensure that every machine, equipment, accessory used is in a safe and efficient order;

d) ensure that each operation, activity is carried on in safe and efficient manner.

8.0. INSPECTION, EXAMINATION, REPAIR AND MAINTENANCE OF HEMM AND OTHER MACHINES:

I. a) A code of practice for inspection, examination and repair of all breakdowns and repairs etc. shall be drawn up and practiced. A copy of the same shall be submitted to this Directorate promptly.

b) Every drilling and earth moving machinery or equipment or accessory (herein after called machine) shall be thoroughly examined at least once in each shift and maintained in good and safe working conditions.

c) A record of all such examinations and maintenances shall be maintained in a bound paged register and shall be signed by the engineer in charge of the shift and countersigned by the mine Engineer.

d) The layout of the workshop for maintenance of HEMM and other equipments shall be as per requirement mentioned in the DGMS Circular No.08 of 2003.

II. a) Every machine shall be allocated at least one day in every week for examination and maintenance, when it shall be thoroughly examined and inspected by the Engineer, who shall satisfy himself that it is mechanically sound and is in efficient working order before it is allowed to be re-used.

b) A report of every maintenance made under clause (a) shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by person making the inspection and countersigned by the Engineer.

c) If the engineer, mechanical Foreman or other competent person making an inspection under Clause (I) (a), notices any defect in any machinery, the said machinery shall not be used until the defect has been remedied.

d) Any defect in machinery reported by its operator shall be promptly attended to.

III. a) Any machinery found to be in an unsafe operating condition shall be tagged at the operator's position; "Out of Service, Do Not Use" and its use shall be prohibited until the unsafe condition has been corrected.

b) All repairs to a machine shall be done at a location, which will provide a safe place for the persons engaged on repairs.

c) Except for testing, trial or adjustment, which must necessarily be done while the machine is in motion, every machine shall be shut down and positive means taken to prevent in operation while any repair or manual lubrication is being done.

d) Any machinery, equipment or part thereof which is suspended or held apart by use of slings, hoists jacks shall be substantially blocked or cribbed before men are permitted to work underneath or between the same.

e) The operator or driver shall not leave the machine while it is running unless it is placed under the charge of a competent person.

f) Power shall be disconnected when repairs are to be carried on any electrically powered machine/ apparatus.

IV. In addition following provisions shall be ensured in general:

1) The approved type of audio visual alarm shall be provided in all equipment.

2) The approved type of fire suppression system shall be provided in all equipment.

3) The stability of HEMM shall be carried out at least once in year and after every major overhaul by an independent agency.

4) The crane and overhead crane shall be subject to proof load test and NDT test once in a year from a competent authority.

5) The pressure vessel receiver are subject to hydraulic and NDT test and shall be carried out by a competent authority.

6) In case of any defect in equipment such as brake, steering and safety device the equipment shall be immediately taken out of use and a record shall be kept.

7) The code of practice for installation operation and maintenance of all equipment shall be prepared and implemented before putting the equipment to use in mine.

8) The safety feature recommend in equipments shall be a part of notice inviting tender for new procurement and the design and drawing shall be obtained from OEM for fitting the same in old equipment.

9.0 PROTECTIVE EQUIPMENT, DUST ETC:

I. PROTECTIVE EQUIPMENT: Every person working in the opencast working shall be provided with, and shall use a hard helmet and protective foot wear of a type approved by the Chief Inspector of Mines.

II. PRECAUTIONS AGAINST DUST: Adequate arrangements, to allay dry dust by wetting shall be made on roads and benches where HEMM, trucks and dumpers operate. All drills shall be provided with wet drilling arrangement and it shall be maintained in working order. No dry drilling operation shall be carried out.

10.0 DRILLING AND BLASTING OF DEEP HOLE:

10.1 GENERAL:

(a) Operations connected with Drilling, Charging, Stemming and Blasting of deep holes shall be placed under overall charge of the Mine Manager or an Assistant Manager, holding at least a second class manager's certificate of competency granted under the Metalliferous Mines Regulations, 1961, who shall supervise the said operations in accordance with the guidelines and directives issued by the Manager.

(b) Notwithstanding anything contained in the Metalliferous Mines Regulations, 1961, preparation of charges, charging and stemming of holes shall be carried out under the personal supervision of a Foreman, who shall fire the shots in deep holes himself.

(c) A proper record for every blast, showing blasting parameters like hole size, spacing, burden, depth of holes, number of holes fired in the round, charge/hole, charge/delay and total charge of explosives fired in the round, with a rough sketch showing the drilling and firing pattern shall be kept maintained in a bound paged register kept for the purpose.

(d) Lives of persons and property not belonging to the management against injury and damages resulting out of blasting or operations being carried on in the mine shall be adequately indemnified. Should there be any controversy in this regard; decision of the DGMS shall be final and binding.

10.2 DRILLS: The following safety features shall provide.

1) Approval type of dust prevention or suppression system.

2) Each moving parts of the machinery shall be guarded/ fenced and also ensure its effectiveness all the time.

3) (i) Emergency of push bottom in

a) Operator's cabin.

b) Main frame.

c) Propeller pendent.

d) Rear end.

(ii) Tripping device to trip the field switch.

4) Thermostat motor protection relay in winding temperature and other related parts.

5) Explosive vent in transformer.

6) Propel interlock (an electric interlock between drilling and propeller operation).

7) High air discharge temperature switch.

8) Low lubrication oil pressure switch.

9) Oil stop valve (electric solenoid valve in compressor lubrication line.

10) No lump circuit.

11) Tower lock and lock check valve.

12) Propel joystick-spring loaded type to return to neutral (dead man safety).

13) Disk-brake and brake valve and its testing parameters.

14) Lock check valve for preventing creeping in drill.

15) Seat belt.

16) Fire resistant hydraulic hoses and wiring near hot zone.

17) Turbo charge guard.

18) Cabin for the operator.

10.3 PRECAUTIONS WHILE DRILLING:

I. The position of every deep hole to be drilled shall be distinctly marked by the mine foreman so as to be readily seen by the drillers.

II. a) No drilling shall be commenced in an area where shots have been fired, until the shot firer has made a thorough examination at all places, including remaining butts of old deep holes, for unexploded charges that the drill may strike.

b) No drill or bore rod or pick shall be inserted in butts of old deep holes even if an examination under clause (a) has failed to reveal presence of explosives.

III. a) Drilling and charging of deep holes shall not be carried out in the same area at the same time.

b) Drilling operations shall not be carried on simultaneously on two benches, at places directly one above the other.

10.4 DUTIES OF DRILL OPERATORS:

I. At the beginning of his shift, the drill operator shall examine the drilling equipment and satisfy himself that:

a) Crown blocks are mounted securely;

b) where compressed air operated drills are used, all hoses' connections are in order;

c) the drilling equipment is in safe working condition; and

d) Wet drilling system is in order.

II. The drill operator shall ensure that:

a) work-persons keep clear of auger and drill stem while the drill is in motion;

b) work-persons do not work under suspended tools, when tools are removed from the holes, these shall be lowered on the platform; and

c) all finished drill holes are properly plugged so as avoid possible leg injuries to any one accidentally stepping into the hole.

10.5 TRANSPSORT OF EXPLOSIVES:

Where explosives are transported in bulk for deep hole blasting, the following precautions shall be taken:

(I) Transport of explosives from the magazine to the priming station or the site of blasting shall not be done except in the original wooden or cardboard packing cases. The quality of explosives transported at any time at the site of blasting shall not exceed the quantity required for use in one round of shots. The explosives shall be transported to the site of blasting not more than 90 minutes before the commencement of charging of the holes.

(II) (a) No mechanically propelled vehicle shall be used for the transport of explosive unless it is of a type approved. A jeep or land rover may be used for the transport of detonators from magazines to priming stations subject to the following conditions -

(i) Not more than 200 detonators are transported in a vehicle at a time.

(ii) The detonators are packed suitably in a wooden box.

(iii) The wooden box containing detonators is placed, inside an outer metal case.

(iv) The outer metal case shall be suitably bolted to the floor of the vehicle of otherwise fixed in a wooden frame so that the container does not move about while the vehicle is in motion and

(v) No person shall ride on the rear portion of the vehicle.

(b) Every vehicle used for the transport of explosive shall be marked or placarded on both sides and ends with the word "EXPLOSIVES" in red letters not less than 15cm high on a white background.

(c) Every mechanically propelled vehicle transporting explosives shall be provided with not less than two fire extinguishers (one suitable for petroleum fire and the other for electrical fire) suitably placed for its convenient use.

(III) (a) The vehicle used for transport of explosives shall not be overloaded and in no case shall the explosive cases be piled higher than the sides of its body.

b) Explosives and detonators shall not be transported the same vehicles, at the same time.

(IV) (a) No person other than the driver and his helper (not below 18 years of age) shall ride on a mechanically propelled vehicle used for the transport of explosives.

b) The explosive van shall not be left unattended.

c) The engine of the vehicle transporting explosives shall be stopped and the brakes set securely before it is unloaded or left standing.

d) A vehicle transporting explosives shall not be driven at a speed exceeding 25km per hour.

e) A vehicle loaded with explosives shall not be taken into repair shop and shall not be parked in a congested place.

f) A vehicle transporting explosives shall not be refueled except in emergencies and then also, when its engine is stopped and other precautions taken to prevent accidents.

g) No trailer shall be attached to a vehicle transporting explosives.

(V) Every vehicle used for the transport of explosives shall be carefully inspected once in every 24 hours by a competent person to ensure that –

i) The fire extinguishers are filled and are in place;

ii) The electric wiring is well insulated and firmly secured;

iii) The chassis engine and body are clean and free from surplus oil and grease;

iv) The fuel tank and feed lines are not leaking; and

v) Lights, brakes and steering mechanism are in good working order.

All reports of every inspection made as above shall be signed and dated by the competent person making the inspection.

(VI) All operations connected with the transportation of explosives shall be conducted under the personal supervision of a Foreman/Blasting in charge so authorised by the Manager and who shall assist the officer in charge of blasting operations at the mine.

(VII) The blasting incharge shall personally search every person engaged in the transport and use of explosives and satisfy himself that no person so engaged has in his possession any cigarette, BEEDI or other smoking apparatus of any kind capable of producing a light, flame or spark.

11.0 DESIGN, OPERATION AND MAINTENANCE OF EXCAVATORS:

(A) DESIGN:

I. a) Every excavator shall be so designed as to afford the operator clear and uninterrupted vision all around.

b) Every excavator shall be provided with -

i) efficient warning devices;

ii) adequate front and rear lights and a portable lamp for emergency, unless the excavator is not used after day-light hours; and

iii) an efficient portable fire extinguisher so placed as to be within easy reach of the operator.

II. A special retracting ladder mounted on the excavating machine.

III. The following safety feature shall be provided with excavators:

i) All functions cut off switch.

ii) Swing motor brake.

iii) Fire resistant hydraulic hoses in place of ordinary hoses to decrease the chance of Fire. All the sleeves and conducts where cables/ wires are passed shall be fire resistant.

iv) Turbo charger guard.

v) Seat belt.

vi) Vent valve on top of hydraulic tank should be able to be removed without any tool.

vii) A baffle plate between cold zone and hot zone.

viii) Provision for limiting of hydraulic cylinders- Stoppers.

(B) OPERATION:

I. Every excavator shall be under the charge of a competent person authorised in writing by the manager, herein called the 'Operator'.

II. a) No person other than the operator or his helper if any, or the manager or any person so authorised in writing by the manager shall ride on any excavator.

b) No person shall be permitted to ride in the bucket of an excavator.

III. Excavator boom shall be lowered to the ground during greasing operations.

IV. a) No excavator shall be operated in a position where any part of the machine suspected and leads or lines are bought closer than 3m to exposed high voltage lines, unless the current has been cut off and positive means taken to prevent the lines from being energized. A notice of this requirement shall be posted at the operator's position.

b) Electrical cables, if any, shall be laid in such a manner that they are not endangered either by falling rocks or by any mobile equipment.

V. The excavator bucket shall be pulled out of the bank as soon as it is full.

VI. a) When being operated in soft or unstable ground, every excavator shall be supported by mats, heavy planks or poles so as to distribute the load of the machinery over larger area and to prevent any danger of the excavator overturning itself.

b) When not in use, the excavator shall be moved to and stood on stable ground.

VII. If more than one stripping machines are in use in any area either on the same bench or on different benches, the machines shall be so spaced that there is no danger of accident from flying or falling objects etc. from one machine to the other.

(C) DUTIES OF EXCAVATOR OPERATORS:

I. a) At the commencement of every shift, the operator shall personally inspect and test the machine, paying special attention to the following details:

i) that every warning device is in working order; or

ii) if the machine is required to work after day-light hours, he shall not take out the machine for work nor shall he work the machine, unless he is satisfied that it is mechanically sound and in efficient working order including the lighting fixtures.

b) The operator shall maintain a record of every inspection made under clause (a) in a bound paged book kept for the purpose and shall sign every entry made therein.

II. a) The operator shall keep the cab window clean so as to ensure clear vision at all times.

b) The walkways in or about the cab of any excavator shall be kept free of loose tools, grease containers or other materials that might fall or present a falling or tripping hazard.

III. a) The operator shall not operate the machine when persons are in such proximity as to be endangered.

b) He shall not swing the bucket over passing haulage unit when the trucks/dumpers are being loaded, he shall swing over the body of the truck/dumpers and not over the cab, unless the cab is protected by a substantially strong cover.

IV. Before leaving the machine, the operator shall lower the bucket to the ground.

V. The operator shall not allow any un-authorized person to ride on the machine.

12.0 DESIGN, OPERATION AND MAINTENANCE OF DUMPERS AND TIPPING TRUCKS:

(A) DESIGN:

(I) Every tipping truck or dumper shall be provided with:

i) efficient brakes;

ii) efficient audible warning devices; and

iii) efficient head-lights, if the truck or dumper is required to work after day-light hours.

iv) an automatic audio-visual alarm which shall operate on reversal and be of such intensity which is not less than 5dB(A) above the surrounding noise level..

(II) In addition, the following safety features shall be provided in dumper:

1. Mechanical steering locking to prevent untoward movement of steering wheel and tyre while work persons working below the cabin while engine is running.

2. Blind spot mirror apart from rear view mirror to enable operator to have clear visibility of blind spot in and around dumpers.

3. Mechanical type Anti-collision device to avoid head to tail collision on haul road such as tail gate, bumper extension or any other strong device.

4. Fire resistant hydraulic hoses in place of ordinary hoses to decrease the chance of fire. All the sleeves and conducts where cables/ wires are passed shall be fire resistant.

5. Seat belt for operator.

6. The maximum speed of vehicle shall be restricted to 30km/hours by blocking higher gear or any other automatic means.

7. Propeller shaft guard.

8. Proximity warning device.

9. Cabin Guard Extension: Canopy shall cover the operator's cabin fully.

10. Exhaust/ Retard Brake: Device to control the speed of truck while operating down the gradient. Refer DGMS (Tech.) Circular No.02 of 2004.

11. Tail gate protection: Protection of operator against collision either by head on or head to tail.

12. Limiting speed device: Enable mine management to decide the maximum speed of vehicle to be operated in mine. The device may be Electronic or mechanical type speed governors.

13. Audio-visual alarm while reversing: The audio-visual alarm provided should confirm to DGMS (Tech.) Circular No.06 of 2020 read with GSR 987(E) of 2018.

14. Provision of two brakes: One of brakes shall be fail safe. For details refer DGMS Circular No.09 of 1999.

15. Body lifting position locking arrangement: A hooter along with an indication is provided to indicate the body is still in lifted position.

16. Fire suppression System: Refer DGMS Circular No.10 of 2004. The fire suppression system shall be a factory fitment and of approved type from this Directorate as per guidelines issued in DGMS(Tech) Circular 06 of 2020.

17. Electric Wires and sleeves are to be of fire resistant quality: To decrease change of fire.

18. Turbo Charge and Exhaust tube coated with heat insulated paint: To decrease change of fire.

19. Battery Cut off Switch: To decrease chance of fire.

20. Retro reflectors on all sides: For visibility of truck during beyond daylight hours.

21. Proximity warning device: To alert operator when approaching any vehicles/obstructions.

22. Rear Vision System: To assist operator during reversing. Refer DGMS Circular No.06 of 2020.

23. Auto Dipping System: To reduce glaring on eyes of operator during night operation.

24. Load Indicator and Recorder: Enables management to detect and prevent over loading.

(B) OPERATION:

I. Every truck or dumper shall be under the charge of a competent person authorised in writing by the manager herein called the 'driver'. No person shall be so authorized unless he possesses a valid driving license to drive a heavy vehicle under the Motor Vehicles Act.

II. a) No person other than the driver or his helper, if any or the manager or any person authorised in writing by the manager shall ride on a truck or dumper.

b) No person shall be permitted to ride on the board of a running truck or dumper.

III. a) As far as possible, loaded trucks or dumpers shall not be reversed on gradients.

b) Sufficient stop blocks shall be provided at every tipping point and these shall be used on every occasion, material is dumped.

c) Standard traffic rules shall be adopted and followed during movement of all trucks and dumpers. They shall be prominently displayed at relevant places in the opencast workings and truck/dumpers roads.

IV. When not in use, every truck or dumper shall be moved to and parked at proper parking places.

V. No person shall be permitted to work on the chassis of a truck or dumper, with the body in a raised position until after the truck or dumper body has been securely blocked in position. The mechanical hoist mechanism shall not be depended upon to hold the body of the truck or dumper in a raised position.

VI. The safety features recommended in the dumpers/ truck/ tippers shall be made a part of the notice inviting tender for new procurement and the design drawing shall be obtained from OEM for fitting the same in old equipment.

(C) DUTIES OF TRUCK/DUMPER OPERATORS:

I. A) Before commencing work, the driver shall personally inspect and test the vehicle, paying special attention to the following details:

i) that the brakes are in working order;

ii) that the warning devices including automatic audio visual reversing alarm are in working order and he shall not take out the vehicle for work nor shall he drive the vehicle, unless he is satisfied that it is mechanically sound and in efficient working order.

b) The driver shall maintain a record of every inspection made under clause (a) in a bound paged book kept for the purpose and shall be signed and dated by the person making the inspection.

II. The driver shall keep the cab window clean so as to ensure clear vision at all times

a) The driver shall ensure that the gear is in neutral position before stopping the engine.

b) The driver shall handle the truck/ dumper carefully and keep it under control at all times. He shall negotiate downhill gradients in low gear so that minimum of braking is required.

c) He shall not drive too fast, shall avoid distractions and shall drive defensively. He shall not attempt to overtake another vehicle unless he can see clearly for enough ahead to be sure that he can pass it safely. He shall also sound audible warning signal before overtaking.

d) When approaching stripping equipment, the driver of the truck or dumper shall sound the audible warning signal and shall not attempt to pass the stripping equipment until he has received a proper audible signal in reply.

e) Before crossing a road or railway line, he shall reduce his speed, look in both directions along the road or line and shall proceed across the road or line only if it is safe to do so.

f) The driver shall not operate the truck or dumper in reverse unless he has a clear view of the area behind the vehicle or he has the assistance of a ‘spotter’ or 'pitmen/ trip man’ duly authorised in writing for the purpose by the manager. He shall give an audible warning signal before reversing a truck or dumper.

g) The driver shall be sure of clearance before driving through tunnels, archways, plant structure etc.

III. The driver shall see that the vehicle is not overloaded and that material is not loaded in a truck or dumper as to project horizontally beyond the sides of its body and that any material projecting beyond the front or rear is indicated by the red flag during day and a red light after daylight hours.

IV. The driver shall not allow any unauthorized person to ride on the vehicle. He shall also not allow more than the authorized number of persons to ride on the vehicle.

(D) TESTING OF BRAKES:

At least once in two weeks, the brakes of the truck, dumper or any such other vehicles shall be tested as indicated below:

I. a) SERVICE BRAKE TEST:

The brake shall be tested on a specified gradient and speed when the vehicle is fully loaded. The vehicle shall stop within a specified distance when the brake is applied. The specified stopping distance shall be obtained from the manufacturer of the vehicle.

b) PARKING BRAKE TEST:

The parking brake shall be capable to hold the vehicle when it is fully loaded and placed at the maximum gradient of roadway, which is permitted for a period of at least ten minutes.

c) A record of such test shall be maintained in a bound paged book and shall be signed by the person carrying out the test and the engineer and manager.

(E) INSPECTION AND OTHER PRECAUTIONS:

I. Surprise inspection shall be carried out by the Engineer/ Superior official at an interval not exceeding 15 days to ensure that the examination schedule of machinery/equipment are carried out.

II. While inflating tyres, suitable protective cages shall be used. Tyres shall never be inflated by sitting either in the front or on the top of the same.

III. While the vehicle is being loaded/ unloaded on gradient, the same shall be secured stationary by parking brake and other means such as suitably designed stopper blocks, which could be placed below the tyres.

IV. Operation and maintenance of heavy machineries such as excavators, dumpers etc. shall be done strictly in accordance with the operation instructions. The operation instructions and maintenance schedule shall be obtained from the manufacturers.

V. a) Automatic fire protection system as mentioned in DGMS (Tech)Circular No. 06 of 2020 shall be provided and maintained in working order for surface heavy machinery such as blast hole drills, hydraulic excavators etc. It is also necessary that the recommended procedure for testing of such fire protection system at a given schedule by the manufacturer be also adopted. The automatic fire systems consist of one or more containers of fire suppressant (usually a dry chemical) connected by a fixed plumbing network to nozzles directed at specific pre-determined fire hazard areas of the machinery.

b) Optical, thermal or any other suitable type of sensing device shall be suitably located to sense the fire and actuate the suppressant to come out through the nozzles already fixed to deal with the fires. An over-riding manual system along with fire alarm shall also be additionally provided for better effectiveness of the system.

VI. Entry of private trucks within the mine premises for dispatch of ore etc. shall be regulated and checks for driving license and its validity, road worthiness of the vehicle, identity of the driver and his helpers, etc. shall be carried out whenever such trucks are allowed to enter into the mine premises. Further crowding of trucks near ore stockyards or inside the mine premises shall not be allowed and only required trucks for immediate loading shall be allowed inside the mine. The management shall provide proper supervision of the trucks and the loading mazdoors. Management shall be responsible for ensuring compliance of all statutory provisions of the Mines Act, 1952 and subordinate legislation framed there under by all the contractor workers/truck drivers, helpers etc in so far as it applies to them. Further no truck shall be allowed to be loaded beyond its permissible capacity.

13.0 It shall be the duty of the Owner, Agent or Manager of every mine to establish and follow a ground control plan for control of all sides, slopes, faces in all opencast workings and overburden dumps. The height of bench and slope of its sides shall be so kept so as to ensure absolute stability under all conditions and shall not, in any case, exceed the general provisions as incorporated in this permission.

Your Faithfully
VIR PRATAP (DIRECTOR - RAIGARH REGION)
THIS IS A SYSTEM GENERATED DOCUMENT, DOES NOT REQUIRE ANY SIGNATURE.